

# RATNESH VANSH SAXENA

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## PROFESSIONAL SUMMARY

DevOps Intern at PearlThoughts and AWS Certified Cloud Practitioner with hands-on experience in AWS cloud, Linux, Docker, Kubernetes, Terraform, Jenkins, and GitHub Actions. Experienced in automating infrastructure, building CI/CD pipelines, and deploying cloud-native applications using modern DevOps practices. Passionate about improving deployment reliability, infrastructure automation, and delivering scalable, secure, and highly available solutions.

## SKILLS

**Cloud:** AWS (EC2, VPC, IAM, S3, RDS, ECR, ECS, EKS, ELB/ALB, Auto Scaling, Route 53, CloudWatch)

**CI/CD & GitOps:** Jenkins, GitHub Actions, Maven, Argo CD, GitOps

**Containers:** Docker, Docker Compose, Kubernetes, Helm

**IaC & Configuration:** Terraform, Ansible

**Security & Monitoring:** SonarQube, Trivy, Prometheus, Grafana, Alertmanager

**Version Control:** Git, GitHub

**Web & Database:** Nginx, Apache Tomcat, MySQL

**Scripting & OS:** Bash, Python, Linux, Ubuntu, Amazon Linux

**Networking:** TCP/IP, DNS, HTTP/HTTPS, SSH, Load Balancing

## PROJECTS

### 3-Tier Architecture Deployment using Terraform

*AWS, EC2, ALB, RDS, VPC, Auto Scaling, Terraform, CloudWatch*

GitHub

- Designed and provisioned a production-grade 3-tier AWS infrastructure using reusable Terraform modules, enabling consistent and repeatable deployments across environments.
- Created a custom VPC with public/private subnets, route tables, Internet Gateway, and NAT Gateway for secure network segmentation.
- Deployed EC2 instances behind an Application Load Balancer with an Auto Scaling Group for traffic distribution and fault tolerance.
- Automated infrastructure provisioning using Terraform with a modular Infrastructure as Code (IaC) approach.
- Configured monitoring and alerting using Amazon CloudWatch metrics, logs, and alarms with end-to-end failover validation.

### TaskFlow Application — GitOps & DevOps Project

*Tech Stack: Docker, Kubernetes, Argo CD, GitHub Actions, AWS*

GitHub

- Developed and containerized a full-stack task management application using React.js, Node.js, and MySQL.
- Implemented a GitOps workflow using Kubernetes and Argo CD, with Git as the single source of truth for automated application deployment and synchronization.
- Built CI/CD pipelines with GitHub Actions and deployed on AWS EC2, reducing manual deployment efforts and ensuring consistent, reliable releases.

### Java Shopping Cart – CI/CD & Production-Style DevOps Deployment

*Technologies: AWS EC2, EKS, Java, Maven, MySQL, Docker, SonarQube, Trivy*

GitHub

- Designed and implemented a complete CI/CD pipeline using Jenkins to automate source code checkout, Maven build, unit testing, SonarQube code analysis, Docker image creation, security scanning, and deployment.
- Containerized the Java-based Shopping Cart application and MySQL database using Docker, ensuring consistent application environments across development and deployment stages.
- Deployed and managed the application on AWS Amazon EKS using Kubernetes Deployments, Services, ConfigMaps, Secrets, and NGINX Ingress and Configured a containerized MySQL database and established secure application.
- Implemented security and quality controls using SonarQube for code quality and Trivy for container vulnerability scanning within the CI/CD pipeline. Improved deployment reliability with an automated Jenkins-to-Kubernetes delivery.

## EXPERIENCE

### DevOps Intern - PearlThoughts - Remote

1-month

- Built and maintained CI/CD pipelines using GitHub Actions and Jenkins to automate deployments.
- Provisioned and updated AWS infrastructure using Terraform, including EC2, VPC, IAM, and networking resources.
- Managed Docker containers and deployed applications to Kubernetes in development and staging environments.
- Monitored applications and servers using Prometheus, Grafana, and CloudWatch, and resolved deployment issues.
- Collaborated with developers to improve deployment reliability and automate operational tasks.

## CERTIFICATIONS

- AWS Certified Cloud Practitioner (2026) – AWS Cloud fundamentals and best practices. Verify
- AWS Academy Cloud Architecting (2025) – Cloud architecture and AWS deployment fundamentals. Verify

## EDUCATION

### Bachelor of Technology (B.Tech), Information Technology

July 2021 – May 2025

Institute of Technology and Management (ITM), Gwalior

CGPA: 7.22/10.0